

MIDTERM EXAMINATION

Coverage: Weeks 1–8 · Foundations, data, labels, linear models, classification, trees, and neural networks

Time allowed: 90 minutes · 40 questions × 2.5 points = 100 points

Instructions

- Choose exactly ONE answer (A–D) per question. No penalty for wrong answers.
- A simple calculator is allowed. All numbers are taken from the lecture hand-calculations.
- Name and student ID on the answer sheet. The answer key begins on a separate page — do not turn to it during the exam.

Part 1 · Foundations & Data (Weeks 1–2)

1. 'Process beats genius' — the Week 1 motto — most directly means:
 - A) Hiring smarter analysts guarantees better returns
 - B) A disciplined, repeatable workflow matters more than one brilliant idea
 - C) Automation should replace all human judgment
 - D) Complex models always beat simple ones
2. In the Week 1 credit-score toy model $z = 2 \cdot \text{Debt} - \text{Income}$, applicant B has Debt = 2.0 and Income = 0.5. B's score z is:
 - A) 1.2
 - B) 1.7
 - C) 3.5
 - D) 4.0
3. Predicting next month's return from labeled historical examples is which type of learning?
 - A) Unsupervised
 - B) Supervised
 - C) Reinforcement
 - D) Self-supervised only
4. Grouping stocks into clusters WITHOUT any labels is:
 - A) Supervised learning
 - B) Unsupervised learning
 - C) Reinforcement learning
 - D) Transfer learning
5. A backtest built only on companies that exist TODAY overstated an average return by 12.4pp in the Week 2 demo. The bias is called:
 - A) Look-ahead bias
 - B) Survivorship bias
 - C) Confirmation bias
 - D) Selection on outcome only
6. Using an earnings figure in a backtest on the day it REFERS to, rather than the (later) day it was PUBLISHED, is:
 - A) Survivorship bias
 - B) Look-ahead bias
 - C) Regime change

D) Feature engineering

7. The Week 2 stance toward any new financial dataset is best summarized as:

- A) Trust it if it comes from a vendor
- B) Data is biased until proven otherwise — audit before modeling
- C) More rows automatically fix quality problems
- D) Cleaning can wait until after model selection

8. Which is a legitimate reason financial ML is HARDER than image classification?

- A) Markets have low noise and stable rules
- B) The signal-to-noise ratio is tiny and the world changes (non-stationarity)
- C) Financial data has no missing values
- D) Labels in finance are always obvious

9. Your model performs brilliantly in backtest but fails live. Weeks 1–2 say the FIRST suspect is:

- A) Bad luck that will reverse
- B) A data or evaluation flaw such as leakage or bias
- C) The market being irrational
- D) Too few neural network layers

10. Why does the course insist on hand calculations before code?

- A) Regulators require handwritten work
- B) You cannot audit a machine whose answer you couldn't reproduce at small scale
- C) Python is unreliable for arithmetic
- D) It makes exams easier to grade

Part 2 • Labels & Linear Models (Weeks 3–4)

11. The triple-barrier method labels a trade using:

- A) Only a fixed time horizon
- B) Profit-take, stop-loss, and time barriers — whichever is hit first
- C) The next day's open price only
- D) Analyst consensus targets

12. In the Week 3 demo, meta-labeling improved precision from 50% to about:

- A) 55%
- B) 60%
- C) 80%
- D) 100%

13. Meta-labeling's second model answers which question?

- A) Which direction will the price move?
- B) Should I ACT on the primary model's signal (bet sizing / filtering)?
- C) What is the fair value of the asset?
- D) Which features should be deleted?

14. Fixed-time labels ('return over the next 20 days') are criticized in Week 3 mainly because they:

- A) Are too hard to compute
- B) Ignore the path — a trade stopped out mid-way gets the wrong label
- C) Require deep learning
- D) Only work for bonds

15. Ridge and Lasso regression both address which problem of plain linear regression?

- A) It cannot use more than 3 features
- B) Overfitting via unstable, oversized coefficients

- C) It cannot output negative values
 - D) It requires labeled data
16. Which statement about Lasso (vs Ridge) is correct?
- A) Lasso can set coefficients exactly to zero, performing feature selection
 - B) Ridge sets coefficients exactly to zero
 - C) Lasso always fits better in-sample
 - D) They are identical penalties
17. The 'factor zoo' (Week 4) refers to:
- A) A database of animal-themed tickers
 - B) Hundreds of published return 'factors', many of which are luck or artifacts
 - C) The set of all neural architectures
 - D) Exchange-listed ETFs
18. A factor is 'statistically significant' in a backtest. Week 4's caution is:
- A) Significance guarantees causation
 - B) With enough factors tried, significance appears by chance — count the trials
 - C) Only t-stats above 1 matter
 - D) Significance implies tradability after costs
19. In the Week 4 panel setup (market + 10 stocks), a stock's market beta is estimated by:
- A) The average of its prices
 - B) Regressing its returns on market returns
 - C) Its dividend yield
 - D) Counting its news mentions
20. Which project decision keeps a Week 4 regression honest?
- A) Using future returns as a feature
 - B) Fitting on one period and evaluating on a LATER, untouched period
 - C) Choosing the model with the best in-sample R^2
 - D) Adding features until R^2 reaches 1

Part 3 • Classification (Weeks 5–6)

21. In the Week 5 logistic demo, an applicant's linear score was $s = 0.6$. Using $\sigma(s) = 1/(1+e^{-s})$ with $e^{-0.6} \approx 0.549$, the default probability is about:
- A) 0.35
 - B) 0.55
 - C) 0.65
 - D) 0.85
22. Logistic regression outputs are best described as:
- A) Class labels only
 - B) Probabilities between 0 and 1
 - C) Unbounded scores
 - D) Distances to the mean
23. A fraud dataset is 99% legitimate. A model that predicts 'legitimate' for everyone scores 99% accuracy. The Week 5 lesson:
- A) The model is excellent
 - B) Accuracy misleads on imbalanced data — use precision/recall/F1
 - C) The dataset is broken
 - D) Switch to a deeper network

24. Precision answers which question?
- A) Of all actual frauds, how many did we catch?
 - B) Of all cases we FLAGGED as fraud, how many really were?
 - C) How fast is the model?
 - D) How many features were used?
25. KNN classifies a new point by:
- A) Fitting a straight line
 - B) A majority vote among its k nearest labeled neighbors
 - C) Growing a decision tree
 - D) Sampling from a prior
26. LOF (Local Outlier Factor) flags a point as anomalous when:
- A) Its value is above the mean
 - B) Its local density is much LOWER than its neighbors' density
 - C) It appears more than once
 - D) Its label is missing
27. The core idea of an SVM is to:
- A) Minimize the number of features
 - B) Find the separating boundary with the LARGEST margin to the nearest points
 - C) Average many weak models
 - D) Cluster the data first
28. The kernel trick lets an SVM:
- A) Run faster on GPUs
 - B) Separate non-linear patterns by implicitly mapping to a higher-dimensional space
 - C) Skip the training step
 - D) Ignore outliers entirely
29. One-Class SVM (OCSVM) performs anomaly detection by:
- A) Voting among neighbors
 - B) Drawing a boundary around 'normal' data and flagging points outside it
 - C) Counting tree splits
 - D) Reconstructing inputs
30. Support vectors are:
- A) All training points equally
 - B) The points closest to the boundary that define the margin
 - C) The features with the largest weights
 - D) The misclassified points only

Part 4 • Trees & Neural Networks (Weeks 7-8)

31. A node's Gini impurity is LOWEST (best) when:
- A) Classes are perfectly mixed 50/50
 - B) The node contains essentially one class
 - C) The node is empty of features
 - D) The tree is deepest
32. Random forests reduce overfitting relative to a single deep tree mainly by:
- A) Using fewer training rows
 - B) Averaging many trees trained on random subsets of data and features
 - C) Removing all randomness
 - D) Pruning to a single split

33. Week 7's caution about impurity-based feature importance (MDI) is that it:
- A) Is always exactly correct
 - B) Can be biased (e.g., toward high-cardinality features) and is not causal
 - C) Only works for regression
 - D) Requires labels to be balanced
34. Isolation Forest detects anomalies because anomalous points:
- A) Need MANY random cuts to isolate
 - B) Need very FEW random cuts to isolate
 - C) Always sit at the sample mean
 - D) Have missing labels
35. XGBoost differs from a random forest in that it:
- A) Builds trees sequentially, each correcting the previous ones' errors
 - B) Builds all trees independently and averages
 - C) Uses no trees at all
 - D) Cannot handle tabular data
36. A single artificial neuron with a sigmoid activation is mathematically equivalent to:
- A) A decision tree
 - B) Logistic regression
 - C) K-Means
 - D) PCA
37. The XOR pattern is important in Week 8 because it:
- A) Can be solved by a single neuron
 - B) Cannot be separated by one linear unit — a hidden layer is required
 - C) Only appears in image data
 - D) Proves networks cannot learn
38. In gradient descent, a learning rate that is far TOO LARGE typically causes:
- A) Very slow but safe convergence
 - B) Divergence or wild oscillation past the minimum
 - C) Automatic regularization
 - D) Exact convergence in one step
39. An autoencoder detects anomalies because:
- A) It memorizes every training point
 - B) Trained on normal data, it reconstructs unusual points POORLY (large error)
 - C) It counts nearest neighbors
 - D) It maximizes the margin
40. Across Weeks 5–8 you met four anomaly detectors (LOF, OCSVM, Isolation Forest, Autoencoder). The practical recommendation was:
- A) Always use only the newest one
 - B) Run several together — each covers the others' blind spots
 - C) Anomaly detection is unnecessary in finance
 - D) Use accuracy to compare them